Southern Nuclear

Sonny Dean
Vice President – Plant Hatch

Hatch Nuclear Plant
11028 Hatch Parkway North
Baxley, GA 31513
912 537 5859 tel
912 366 2077 fax

December 10, 2020

Docket Nos.: 50-321

NL-20-1366

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant Unit 1
Licensee Event Report 2020-001-00
1A EDG Inoperable Due to Water in the Day Tank

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins at 912.453.2342.

Respectfully submitted,

Sonny Dean
Vice President – Hatch

SD/CJC/SCM

Enclosure: LER 2020-001-00

cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 1

Licensee Event Report 2020-001-00

1A EDG Inoperable Due to Water in the Day Tank

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Edwin I. Hatch Nuclear Plant - Unit 1	05000 321	1 OF 2

4. Title
1A EDG inoperable due to water in day tank

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number 05000
12	03	2020	2020	- 001 -	00	12	10	2020	Facility Name	Docket Number 05000

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Carl James Collins - Licensing Manager	912-453-2342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	EK	DG	F010	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On 5/8/20 at 1714, while taking a fuel sample from the 1A Emergency Diesel Generator (EDG) day tank, 14.73 gallons of water was found in the tank. The water was immediately drained from the day tank. Additionally, from 5/14/20 through 5/16/20, the day tank and the main fuel oil storage tank (FOST) were drained, cleaned, inspected and returned to service. A past operability review was conducted and on 12/03/20 the results concluded that the EDG would not have been able to perform its safety function with the amount of water found in the day tank.

The source of water in the day tank was from accumulated water in the FOST being pumped to the day tank during a fuel transfer system surveillance on 4/29/20. Water from an underground pipe leak had entered a nearby pull box, traversed conduit connecting to the FOST manway, collected in the manway and then seeped into the FOST through threaded connections on top of the tank.

The direct cause of the event was clogging of the FOST manway drains which allowed water to collect in the manway. There was no maintenance strategy to inspect and clean the manway drains. This was corrected by cleaning debris from the drains and implementing a maintenance strategy for periodic cleaning.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant - Unit 1	05000- 321	2020	001	00

NARRATIVE

Event Description:

On 5/7/20, Unit 1 was in MODE 1 at 100 percent rated thermal power. While performing an Emergency Diesel Generator (EDG) fuel oil transfer pump surveillance, standing water was observed in the manway of the 1A EDG Fuel Oil Storage Tank (FOST). Due to the standing water on the FOST, a sample of the fuel was taken and analyzed. The sample found water in the diesel fuel. Subsequently, on 5/8/20 at 1714, the associated 1A EDG day tank was sampled and 14.73 gallons of water was found. Actions were taken to immediately drain the water from the day tank. Between 5/14/20 and 5/16/20 the day tank and FOST were drained, cleaned, inspected and returned to service.

Analysis and testing were conducted in order to determine if the EDG was capable of performing its safety function with the quantity of water found in the day tank. It was ultimately concluded on 12/03/20 that the EDG would not have been able to perform its safety function. Since the EDG was last successfully run on 4/29/20 and water was transferred from the FOST during this surveillance, it is likely the EDG would not have been able to perform its safety function beginning on 4/29/20 once the water settled in the day tank after the EDG surveillance. The time period between 4/29/20 and when the water was drained on 5/8/20 exceeds the Technical Specification (TS) Completion Time of 72 hours, thus making this issue reportable as a condition prohibited by TS per 10CFR50.73(a)(2)(i)(B).

Event Cause Analysis:

Causal analysis was performed to determine the source of the water in the day tank. It concluded that standing water on top of the 1A EDG FOST had seeped into the FOST through threaded connectors on top of the tank. The water then settled to the bottom of the FOST and was pumped into the day tank during the EDG surveillance on 4/29/20. The source of the standing water on top of the FOST was from an underground pipe leak that entered a nearby pull box and then flowed through conduit connecting to the FOST manway. It was found that drain holes in the FOST manway were clogged, allowing water to collect in the manway. There was no maintenance strategy to inspect and clean the manway drain holes to prevent clogging. It was also determined that the sump pump in the pull box was non-functional and that conduit seals between the pull box and the FOST manway were degraded. The non-functional sump pump and degraded conduit seals contributed to water collecting in the pull box and migrating to the FOST manway.

Safety Assessment:

There were no safety consequences as a result of this event. While the 1A EDG was not operable, both the 1B and 1C EDG's remained operable on Unit 1. However, due to the 1A EDG being inoperable longer than allowed by TS, this event is reportable as a condition prohibited by TS per 10CFR50.73(a)(2)(i)(B).

Corrective Actions:

The FOST manway drains were cleaned. Additionally, a maintenance strategy for cleaning manway drains periodically was implemented. The pull box conduit seals, sump pump and the underground pipe leak have been repaired.

Previous Similar Events:

None.